Grep Me Out!

Grep is one among the systems administrator's 'Swiss Army knife' set of tools, and is extremely useful to search for strings and patterns in a group of files, or even sub-folders. This article introduces the basics of Grep, provides examples of advanced use and links you to further reading.

Grep (an acronym for 'Global Regular Expression Print') is installed by default on almost every distribution of Linux, BSD and UNIX, and is even available for Windows. GNU and the Free Software Foundation distribute Grep as part of their suite of open source tools (the official website is http://www.gnu.org/software/Grep/). This tutorial focuses primarily on this GNU version, as it is currently the most widely used.

Grep finds a string in a given file or input, quickly and efficiently. While most everyday uses of the command are simple, there are a variety of more advanced uses that most people don't know about—including regular expressions and more, which can become quite complicated.

The tool has its roots in an extended regular expression syntax that was added to UNIX after Ken Thompson's original regular expression implementation. The latter searches for any of a list of fixed strings, using the Aho-Corasick algorithm. These variants are embodied in most modern Grep implementations as command-line switches (and standardised as *-E* and *-F* in POSIX.2). In such combined implementations, Grep may also behave differently depending on the name by which it is invoked, allowing *fGrep*, *eGrep*, and *Grep* to be links to the same program.

There are two ways to provide input to Grep, each with its own particular uses. First, Grep can be used to search a given file or files on a system (including a recursive search through sub-folders). Grep also accepts inputs (usually via a pipe) from another command or series of commands.

Regular expressions

A regular expression, often shortened to 'regex' or 'regexp', is a way of specifying a pattern (a particular set of characters or words) in text that can be applied to variable inputs to find all occurrences that match the pattern. Regexes enhance the ability to meaningfully process text content, especially when combined with other commands.

Usually, regular expressions are included in the Grep command in the following format:

```
Grep [options] [regexp] [filename]
```

GNU Grep uses the GNU version of regular expressions, which is very similar (but not identical) to POSIX regular expressions. In fact, most varieties of regular expressions are quite